

IBM WATSONX CHALLENGE — USE CASE 2

Infrastructure Incident Triage Assistant

RAG-powered AI that triages production incidents in under 3 seconds

IBM Bob

RAG

FastAPI

Gemini 2.0 Flash

Groq Backup

THE PROBLEM

"When an alert fires at 2 AM, on-call engineers spend 30–45 minutes just gathering context before they can start fixing anything."

Documentation Scatter

Runbooks in Confluence, incidents in Jira, contacts in PagerDuty — all separate logins, all under pressure.

Knowledge Silos

Resolution knowledge lives in senior engineers' heads — unavailable at 2 AM and impossible to scale.

Escalation Delays

Without fast context, engineers escalate too early or too late — wrong person paged, SLA breached, customer impact grows.

THE SOLUTION

A single-page AI assistant that ingests all operational documentation and answers any alert in plain English — grounded entirely in your own runbooks and incident history.



WHY RAG — NOT A GENERIC CHATBOT

GENERIC CHATBOT	THIS ASSISTANT
Makes up runbook steps	Only uses steps from your actual runbooks
No incident history	Surfaces INC-XXXX with root cause + resolution time
No source citations	Every answer cites the exact document used
Stale training data	Upload a runbook → indexed in seconds, queryable immediately
No conversation memory	Last 3 exchanges sent with every query — follow-ups work

BUSINESS IMPACT

45 min

↓

< 30 sec

Time to first triage action

2–3×

↓

0–1×

Senior engineer interruptions per incident

30–45 min

↓

< 10 sec

Post-mortem report generation

KPI	BEFORE	AFTER
MTTR (Mean Time to Resolution)	45–90 min	15–30 min
Runbook retrieval	15–20 min (manual search)	< 2 seconds
Senior engineer interruptions	2–3 per incident	0–1 per incident
Time to escalation path	5–10 min	< 5 seconds, cited from policy
Server startup (after first run)	N/A	< 5 seconds (index loaded from disk)

FEATURE SET

- **Natural language triage** — any alert in plain English
 - **Streaming responses** — word-by-word via SSE
 - **Severity auto-detection** — P1/P2/P3/P4 banner
 - **Diagnostic commands** — copy-ready bash, kubectl, SQL
 - **Past incident surfacing** — INC-XXXX with root cause
 - **Escalation guidance** — who to page, when, which channel
- **Post-mortem generator** — full report in 10 seconds
 - **Live document upload** — drag-and-drop, re-indexed instantly
 - **Chat session history** — saved, scrollable, replayable
 - **Similar session surfacing** — related past incidents shown
 - **LLM fallback** — Groq auto-activates if Gemini rate-limits
 - **Persistent vector index** — loads from disk in < 200ms

CURRENT KNOWLEDGE BASE

TYPE	COUNT	COVERAGE
Runbooks	5	High CPU, Disk Space, Service Down, Network Latency, Permission Drift
Incident Post-Mortems	5	INC-2847, INC-3012, INC-3301, INC-3455, INC-3601
Policy & Reference	2	Escalation Policy (P1–P4), Alert Definitions (30 alerts)
Total indexed	12	351 chunks · 384-dim embeddings

TECHNOLOGY STACK

Backend

FastAPI — async Python API server with SSE streaming

Sentence Transformers — all-MiniLM-L6-v2, local, no API key

NumPy + JSON — persistent vector index, zero infrastructure

Gemini 2.0 Flash — primary LLM (free tier)

Groq / Llama 3.1 — automatic backup LLM on rate limit

Frontend

Single HTML file — no React, no npm, no build step

SSE streaming — tokens rendered word-by-word as they arrive

localStorage — chat session history, survives page refresh

Drag-and-drop upload — live re-index via POST /upload

Markdown renderer — code blocks, tables, copy button

IBM BOB ALIGNMENT

IBM BOB PRINCIPLE	IMPLEMENTATION
Grounded responses	Every answer uses only retrieved document chunks — never LLM training data alone
Cited answers	Source chips on every response link back to the exact document and folder
Actionable output	Structured: Immediate Actions → Diagnostic Commands → Root Causes → Escalation
Contextual awareness	Last 3 conversation exchanges injected into every prompt for multi-turn reasoning
Granite-ready	Single function swap — replace Gemini with ibm/granite-3-8b-instruct for on-premise

ROADMAP — FUTURE ENHANCEMENTS

- 1

Save from Chat

Type "save this as an incident" — writes a formatted post-mortem .md to incidents/ and live re-indexes it. KB grows automatically during real incidents.
- 2

Confidence Score Filter

Warning banner when cosine similarity < 0.4 — tells engineers the assistant has no grounded runbook for this query. Prevents hallucinations in live incidents.
- 3

Runbook Gap Detector

GET /gaps — compares all 30 alert definitions against indexed runbooks and returns the exact list of alerts with no coverage.
- 4

Server-Side Chat History + Vector Search

SQLite for session storage + ChromaDB for semantic similar-session search across the full team. MiniLM already loaded — zero new dependencies.
- 5

Incremental Re-Index

Per-file fingerprints — only new or changed documents re-embedded on upload. Keeps live uploads instant as knowledge base scales.
- 6

IBM Granite Swap-In + Full On-Premise

Replace Gemini with ibm/granite-3-8b-instruct. One function change — entire stack becomes air-gapped with zero external API calls.

< 3s

Alert triaged, steps surfaced

12

Documents indexed, 351 chunks

0

External infrastructure required